



AI-powered genomic trait prediction and dataset comparison

Upload Dataset A (required)



AncestryDNA_raw.txt ✕
17.5MB



Upload Dataset B (optional)



anonymo...37_raw.txt ✕
15.3MB



✓ Processing completed

Dataset A

✓ Profiled — 5/5 traits evaluated

Dataset B

✓ Profiled — 5/5 traits evaluated

Dataset Quality

Dataset A Dataset B

Rows processed

677,436

Encoding

UTF-8

Genotype layout

Two-colu...

Detected findings

Methodology caveats

2

9

Findings detail

- `duplicate_chr_pos_check`: Rows whose (chromosome, position) pair (at field positions 1, 2) occurs in two or more rows: 614 duplicated pair(s) found. (1270)
- `missing_value_scanner`: Literal value frequencies observed across 2 designated column(s): 7 distinct value(s) found. (1354872)

Open questions

- Whether purely numeric chromosome labels (as observed in Dataset A) and numeric-plus-alphabetic chromosome labels (as observed in Dataset B) are equivalent or directly comparable across files has not been established. Any future process that joins, merges, or compares records by chromosome across files of these differing kinds risks misalignment if this is not explicitly reconciled against an authoritative reference.
- The biological meaning of non-standard numeric chromosome codes (i.e. codes beyond the conventional 1-22 autosome range) observed in a file's chromosome labels is not defined within the file itself. Interpreting these codes without an authoritative source risks silently mislabeling sex-chromosome or mitochondrial data.
- Whether missing/no-call tokens are handled equivalently across files with differing column layouts and literal token conventions has not been established. A future process that hard-codes one file's missing-value convention risks silently failing to recognize missing data expressed under a different convention.
- Whether duplicate (chromosome, position) pairs observed within a file reflect probe redesigns, multi-allelic sites, re-genotyping, or data artefacts has not been determined. Treating them as errors, or as valid multi-allelic entries, without further investigation risks either discarding valid data or retaining erroneous data.

- Whether a two-column allele-pair genotype layout and a single-column genotype-string layout (including haploid/indel variants) can be unified into a common representation without loss or misrepresentation has not been established. Any future unification of these encodings carries a risk of information loss if the haploid/diploid/indel distinctions are not preserved.
- Each file's header claims a forward/plus-strand orientation, but no independent verification of strand consistency has been performed. Any future position-based operation relying on strand consistency carries a risk of silent error if this claim does not hold in practice.
- Whether a file's content is decodable only under certain text encodings (e.g. not as pure ASCII) has an identified source and location has not been established. This warrants further investigation before any text-processing logic assumes uniform character encoding across files.
- Genotype data is intrinsically personal and can in principle be re-identified even with direct identifiers removed. Any handling of files of this kind inherits a data-protection risk that must be addressed by a dedicated, future security/privacy design, not assumed away here.
- Whether every value in a file's RSID column is a standard dbSNP identifier, as opposed to a platform-internal identifier, has not been established. A future process that assumes all RSID-column values are standard dbSNP identifiers risks mishandling or misinterpreting the platform-internal subset.

Trait Predictions

Dataset A Dataset B

✓  Lactase Persistence

lactase non-persistent

HIGH



Category-based confidence — not a numeric probability.

 Evidence

Supporting SNPs: rs4988235

- rs4988235 = GG

 References

LCT/MCM6 lactase persistence variant (Enattah et al., 2002).

GRCh37 forward-strand locus NC_000002.11:g.136608646 (position-matched NCBI/ClinVar record); historically cited as -13910C>T on the gene-relative strand (Phase 2 Architecture Blueprint, Section 9.3).

 Limitations

Population-generalizability caveat applies to this trait — see Methodology & Limitations below.

▼ Earwax Type

wet earwax

HIGH



Category-based confidence — not a numeric probability.

 Evidence

Supporting SNPs: rs17822931

- rs17822931 = CC

 References

ABCC11 earwax association (Yoshiura et al., 2006).

GRCh37 forward-strand locus NC_000016.9:g.48258198 (position-matched ClinVar record, Chr16:48258198); historically cited as 538G>A on the gene coding strand (Phase 2 Architecture Blueprint, Section 9.3).

 Limitations

Population-generalizability caveat applies to this trait — see Methodology & Limitations below.

▼  ACTN3 (Alpha-Actinin-3)

ACTN3 RR (power-associated, alpha-actinin-3 present)

MODERATE



Category-based confidence — not a numeric probability.

 Evidence

Supporting SNPs: rs1815739

- rs1815739 = CC

 References

ACTN3 R577X variant and athletic performance association (Yang et al., 2003).

Checked against the standard C/T allele convention used across all sources reviewed; no strand conversion required (Phase 2 Architecture Blueprint, Section 9.3).

 Limitations

Population-generalizability caveat applies to this trait — see Methodology & Limitations below.

▼ Bitter Taste (TAS2R38)

non-taster (AVI/AVI)

HIGH



Category-based confidence — not a numeric probability.

 Evidence

Supporting SNPs: 3 SNPs

- rs713598 = CC
- rs1726866 = AA
- rs10246939 = TT

 References

TAS2R38 PAV/AVI bitter taste haplotypes (Kim et al., 2003).

Consumer genotype data lacks phase information; diplotype interpretation uses common haplotype assumptions.

GRCh37 forward-strand locus NC_000007.13:g.141673345, verified directly against the dbSNP RefSNP report and its linked ClinVar record (RCV000003038.1, 'Phenylthiocarbamide tasting', allele G) — resolving the one open item from

Phase 2 Architecture Blueprint Section 20, Item #1. Note: one secondary source (SNPedia) states the opposite allele-to-phenotype direction for this SNP; this entry follows the primary dbSNP/ClinVar record, corroborated by an independent peer-reviewed source, over that secondary source — flagged here for visibility, not silently resolved.

GRCh37 forward-strand locus NC_000007.13:g.141672705, verified directly against the dbSNP RefSNP report and its linked ClinVar record (RCV000003039.1, 'Phenylthiocarbamide tasting', allele G) — resolving the one open item from Phase 2 Architecture Blueprint Section 20, Item #1.

GRCh37 forward-strand locus NC_000007.13:g.141672604, verified directly against the dbSNP RefSNP report and its linked ClinVar record (RCV000003040.1, 'Phenylthiocarbamide tasting', allele C) — resolving the one open item from Phase 2 Architecture Blueprint Section 20, Item #1.

 Limitations

Population-generalizability caveat applies to this trait — see Methodology & Limitations below.

▼ Eye Colour (IrisPlex)

Blue: 85.2%, Other: 9.5%, Brown: 5.3%. Most likely category: Blue.

HIGH



Category-based confidence — not a numeric probability.

 Evidence

Supporting SNPs: 6 SNPs

- rs12913832 = GG
- rs1800407 = CC
- rs12896399 = GG

- rs16891982 = GG
- rs1393350 = GG
- rs12203592 = CC

 References

IrisPlex 6-SNP panel and eye colour prediction model (Walsh et al., 2011).

Walsh et al. (2011) IrisPlex eye-colour prediction model; variant information cross-checked with dbSNP/ClinVar.

Andersen et al. (2016) eye pigmentation study; allele representation verified using dbSNP/ClinVar records.

Walsh et al. (2011) IrisPlex model; SLC24A4 pigmentation variant information verified with dbSNP.

Walsh et al. (2011) IrisPlex model; SLC45A2 pigmentation association verified with dbSNP records.

Andersen et al. (2016) eye pigmentation study; TYR variant association cross-checked with dbSNP.

IrisPlex eye-colour model (Walsh et al., 2011); IRF4 pigmentation association reported in pigmentation studies.

 Limitations

Population-generalizability caveat applies to this trait — see Methodology & Limitations below.

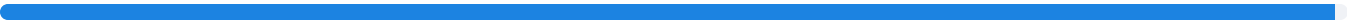
▼ Trait Summary

Dataset A Dataset B

Trait	Prediction	Confidence	Supporting SNPs
Lactase Persistence	lactase non-persistent	HIGH	rs4988235
Earwax Type	wet earwax	HIGH	rs17822931
ACTN3 (Alpha-Actinin-3)	ACTN3 RR (power-associated, alpha-actinin-3 present)	MODERATE	rs1815739
Bitter Taste (TAS2R38)	non-taster (AVI/AVI)	HIGH	3 SNPs
Eye Colour (IrisPlex)	Blue: 85.2%, Other: 9.5%, Brown: 5.3%. Most likely category: Blue.	HIGH	6 SNPs

Dataset Comparison

Comparable genotype agreement



99.99%

Matching SNPs

155,414

Conflicting SNPs

8

Platform coverage differences

Platform coverage differences are not biological disagreement.

Structural mismatches

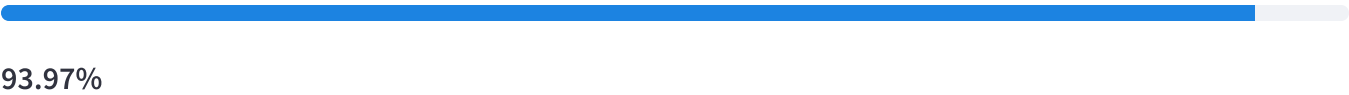
9,423

Missing / no-call

548

Overall shared-marker agreement

Across all shared SNPs, including no-call and structural differences.



Identity likelihood: low concordance

Rigorous identity or relatedness determination uses identity-by-descent/identity-by-state estimation across many more markers than this concordance check performs.

Per-trait comparison

Trait	Dataset A	Dataset B	Agreement
ACTN3 (Alpha-Actinin-3)	ACTN3 RR (power-associated, alpha-actinin-3 present)	ACTN3 RR (power-associated, alpha-actinin-3 present)	✓ Agree
Bitter Taste (TAS2R38)	non-taster (AVI/AVI)	non-taster (AVI/AVI)	✓ Agree
Earwax Type	wet earwax	wet earwax	✓ Agree
Eye Colour (IrisPlex)	Blue: 85.2%, Other: 9.5%, Brown: 5.3%. Most likely category: Blue.	Blue: 85.2%, Other: 9.5%, Brown: 5.3%. Most likely category: Blue.	✓ Agree
Lactase Persistence	lactase non-persistent	lactase non-persistent	✓ Agree

▼ ⚠ Methodology and Limitations

Ancestry generalizability

Applies to every trait prediction above: Most trait associations were established primarily in European-ancestry cohorts and may not generalize equally across populations.

Identity/relatedness methodology

See the Dataset Comparison section above.

 Download report (JSON)